

Customer Analytics on the Olist E-Commerce Dataset

This project presents an end-to-end customer analytics solution using the **Olist Brazilian E-Commerce Public Dataset**, one of the largest publicly available e-commerce datasets. The analysis combines exploratory data analysis, customer segmentation, predictive modeling, and business intelligence techniques to generate actionable insights for improving customer experience, operational efficiency, and business performance.

96,096 Unique Customers	R\$15.8M Total Revenue	4.08 / 5 Avg. Review Score	3.1% Repeat Customer Rate
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Executive Summary

Olist connects small Brazilian merchants to major online marketplaces. This project examines two years of transactional, customer, product, and review data to uncover insights into customer value, satisfaction, delivery performance, and retention. The platform generated R\$15.8 million in revenue from approximately 96,000 customers, while maintaining a strong average review score of 4.08/5. Yet the analysis uncovered a critical business insight: only 3.1% of customers placed a second order. This finding shifts the strategic focus from customer acquisition to customer retention, suggesting that increasing repeat purchases even by a small margin could deliver substantial gains in revenue and customer lifetime value.

Customer Segmentation

RFM Segmentation (Recency, Frequency, Monetary)

Customers were scored and grouped into five behavioral segments. Frequency barely differentiates segments because most Olist customers buy only once — recency and monetary value carry the signal.

Segment	Avg. Recency (days)	Avg. Frequency	Avg. Monetary (R\$)
Champions	140	1.10	175.24 (Highest)
Loyal Customers	219	~1.0	161.87
Potential Loyalists	140	1.00	164 (Moderate)
At Risk	446	~1.0	188.03
Lost Customers	353	1.00	108 (Lowest)

K-Means Behavioral Clusters (k = 4)

Customers were grouped into five behavioral segments using RFM analysis. However, with most customers making only a single purchase, **frequency offered little separation between segments**, while **recency and monetary value became the dominant indicators of customer behavior and value**.

Cluster	Avg. Spend	Orders	Avg. Review	Delivery
Dissatisfied one-time Customers	\$151.21	1	1.70	21.6 days
Low-Spending One-Time Customers	\$128	1	4.66	10.5 days
High-Value Customers	\$1,146.53	~1	3.98	14 days
Loyal Repeat Customers	\$287.28	2.12	4.12	12 days

The **Dissatisfied** segment represents a clear operational challenge rather than a marketing issue. With an average delivery time of **21.6 days** approximately twice the platform average and the lowest average review score (**1.71**), improving delivery speed for this group is likely the most impactful action for increasing customer satisfaction and retention.

Predictive Modeling

Four supervised models were built on engineered customer- and order-level features, each comparing a linear/logistic baseline against a Random Forest.

Task	Model	Key Metric(s)
Customer Lifetime Value (regression)	Linear Regression	R-squared 0.758 · MAE 0.013
	Random Forest	R-squared 0.973 · MAE 0.001
Customer Churn (90+ days inactive)	Logistic Regression	Accuracy 73.0% · ROC-AUC 0.798
	Random Forest	Accuracy 93.1% · ROC-AUC 0.911
Late Delivery (classification)	Random Forest	Accuracy 93.6% · ROC-AUC 0.776
Low Review Score (classification)	Logistic Regression	Accuracy 70.0% · ROC-AUC 0.688
	Random Forest	Accuracy ~90%+ (see notebook 16)

Across all predictive tasks, the **Random Forest** model consistently outperformed the linear baseline, with the largest improvement observed in **Customer Lifetime Value (CLV) prediction ($R^2 = 0.97$ vs. 0.76)**. Feature importance analysis revealed that **category diversity**, rather than total spending, was the strongest predictor of CLV, indicating that customers purchasing across multiple product categories represent the highest long-term value.

Delivery & Satisfaction Drivers

- Delivery performance is the primary driver of both customer churn and low review scores. Delivery delay and overall delivery time consistently rank above price, freight charges, and payment behavior in predicting negative customer outcomes.
- Most orders are delivered within one to two weeks and before the estimated delivery date. However, a small proportion of significantly delayed shipments creates a disproportionate share of customer dissatisfaction and poor reviews.
- Delivery performance varies across both product categories and geographic regions. Bulky products, such as furniture, require longer delivery times, while customers in remote states consistently experience slower deliveries than those in major metropolitan areas.
- Higher freight charges do not consistently translate into faster deliveries. This suggests that logistics network efficiency including shipping distance, carrier routing, and operational constraints—has a greater impact on delivery speed than shipping costs alone.

Geographic & Product Category Insights

- Customers and revenue are concentrated in a small number of markets. Consistent with the Pareto principle, approximately 45% of customers generate 80% of total revenue, with the majority of high-value customers located in just six states, highlighting key markets for retention and targeted marketing.
- Bed, Bath & Table leads in both sales volume and revenue. In contrast, Computers and Watches & Gifts achieve strong revenue despite lower order volumes, reflecting their higher average selling prices and greater revenue per transaction.
- Office Furniture is the weakest-performing major category. It records the highest average freight cost alongside the lowest average customer review score, making it a clear priority for improvements in logistics, shipping efficiency, and product quality.

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- Health & Beauty and Sports & Leisure combine high revenue with consistently strong customer satisfaction, making them the most attractive categories for additional marketing investment and future growth initiatives.
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Recommendations

- Prioritize delivery improvements for the Dissatisfied segment. With an average delivery time of 21.6 days and the lowest average review score (1.71/5), this group represents the greatest opportunity for operational improvement. Addressing delivery delays is likely to have a larger impact on customer satisfaction and retention than additional marketing efforts.
 - Focus retention campaigns on At-Risk customers rather than Lost customers. At-Risk customers have already demonstrated relatively high spending (average R\$188 per customer), making them more cost-effective to re-engage than acquiring new customers with similar lifetime value.
 - Leverage the Random Forest CLV model to optimize retention spending. Since category diversity is the strongest predictor of customer lifetime value, cross-selling strategies that encourage customers to purchase from a second product category may generate greater long-term value than campaigns aimed solely at increasing repeat purchases.
 - Improve fulfilment performance in the Office Furniture category. Despite strong demand, high freight costs and poor delivery experiences are limiting customer satisfaction and revenue potential. Investments in logistics and fulfilment efficiency should take precedence over increasing marketing spend for this category.
 - Invest in logistics across the six highest-revenue states. These markets account for the majority of customers and revenue, so improving delivery speed and service quality in these regions is expected to produce the greatest return on investment through higher customer satisfaction, repeat purchases, and overall revenue growth.
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Methodology & Tools

- The analysis uses the public Olist Brazilian E-Commerce dataset (2016–2018), comprising customer, order, order item, payment, review, product, seller, and geolocation data. The data pipeline was built using DuckDB and pandas for data ingestion, cleaning, integration, and feature engineering. Exploratory data analysis and visualizations were created using matplotlib and seaborn, while scikit-learn was used to implement K-Means clustering, Linear Regression, Logistic Regression, and Random Forest models.
- The project is organized into 16 sequential Jupyter notebooks, covering the complete customer analytics workflow: data loading and preprocessing, feature engineering, exploratory data analysis, geographic and product category analysis, delivery performance, customer satisfaction analysis, Pareto (80/20) analysis, RFM segmentation, Customer Lifetime Value (CLV) analysis, K-Means customer segmentation, and four predictive modeling tasks CLV prediction, repeat customer prediction, late delivery prediction, and low review score prediction.
- This end-to-end pipeline demonstrates how transactional e-commerce data can be transformed into actionable business insights through descriptive, diagnostic, predictive, and prescriptive analytics.

Full analysis, code, and visualizations available in the accompanying notebooks (01–16).